



LIMIT TEST FOR SOME METALS AND NON METALS

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INTRODUCTION

- ❑ Limit tests for metals and non-metals are essential analytical procedures used to detect and quantify trace amounts of impurities in substances, ensuring they meet specified standards.
- ❑ These tests are particularly important in pharmaceuticals, food, and environmental analysis to ensure safety and compliance.

METALS LIMIT TEST

Limit test for Lead

- ❑ Definition:- Its the test designed to know the allowable amount of lead atom presenting in the given sample (pharmaceutical substance).
- ❑ Lead impurity is the most undesirable impurity in medical compound that may caused by the leachable property of the container.
- ❑ Some Lead impurity issues on medical or nutritional substances can damage the developing brain of infants ,leads to learning disabilities, bring behavioral problems and etc.

PRINCIPLES

- ❑ This test relies on the physical properties of the lead, which is its tendency to form color when treated with specific agent which is diphenylthiocarbazone **the reaction between lead and dithizone results in formation of red color.**
- ❑ However, there are metals like Mercury and Cadmium that can form red color the same way as lead when reacted with dithizone, here the lack of specificity seems to be controlled as it may cause false positive. The techniques like spectrophotometry and extraction methods are used to solve this problem.

Procedure

Test Sample	Standard Compound
A known quantity of sample solution is transferred in a separating funnel	A standard lead solution is prepared equivalent to the amount of lead permitted in the sample under examination
Add 6ml of ammonium citrate	Add 6ml of ammonium citrate
Add 2 ml of potassium cyanide and 2 ml of hydroxylamine hydrochloride	Add 2 ml of potassium cyanide and 2 ml of hydroxylamine hydrochloride
Add 2 drops of phenol red	Add 2 drops of phenol red
Make solution alkaline by adding ammonia solution.	Make solution alkaline by adding ammonia solution
Extract with 5 ml of dithizone until it becomes green	Extract with 5 ml of dithizone until it becomes green
Combine dithizone extracts are shaken for 30 mins with 30 ml of nitric acid and the chloroform layer is discarded	Combine dithizone extracts are shaken for 30 mins with 30 ml of nitric acid and the chloroform layer is discarded
To the acid solution add 5 ml of standard dithizone solution	To the acid solution add 5 ml of standard dithizone solution
Add 4 ml of ammonium cyanide	Add 4 ml of ammonium cyanide
Shake for 30 mins	Shake for 30 mins
Observe the color	Observe the color

□ In this method, the lead present as impurity is separated by extraction method, this is employed in order to increase the validity of test. This separation method relies on the pH adjustment by ammonium citrate and potassium cyanide.

Reasons

- ❑ **Ammonium citrate** - Acts as a buffering agent .
- ❑ **Potassium cyanide** - masks the metals like mercury and cadmium.
- ❑ **Phenol red** in this test is used as indicator.

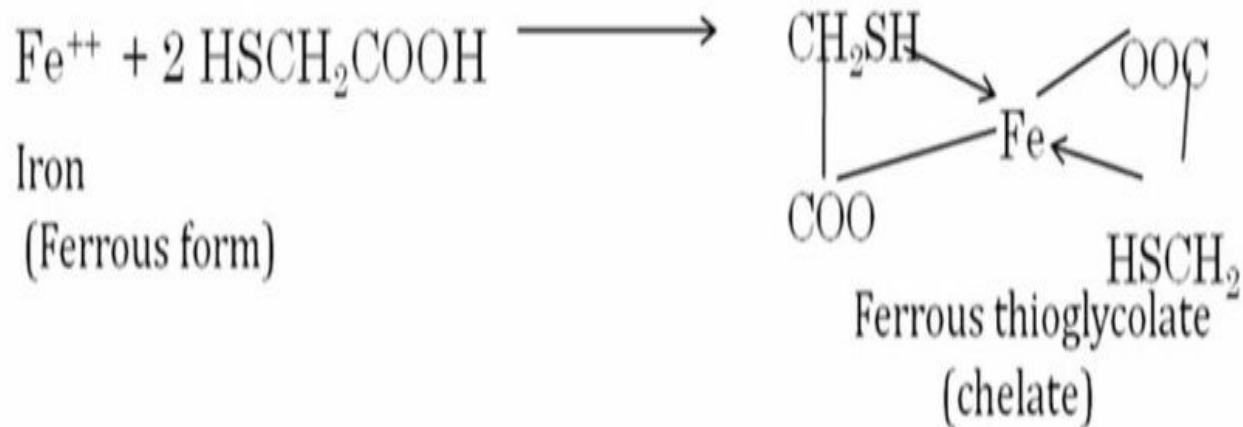
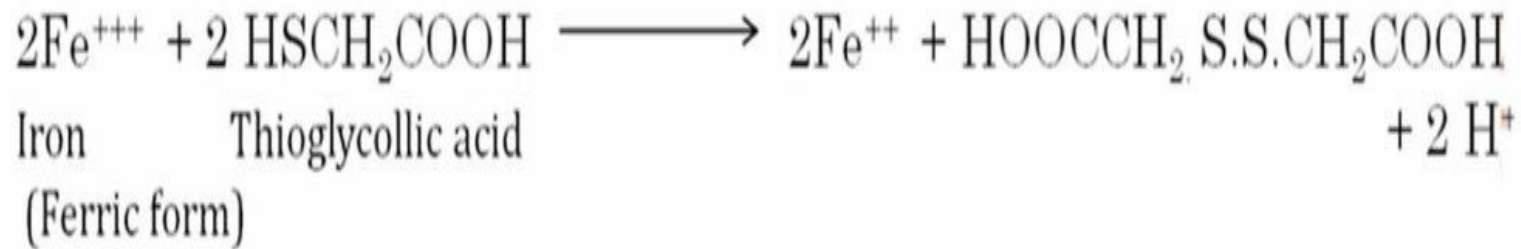
Observation and Comparison

- ❑ The red color is observed .
- ❑ But the redness may vary depending on the lead impurity level.
- ❑ If the redness of the color observed is beyond the standard solution the pharmaceutical formulation fails ,has safety issue and the vice versa . Pharmacopeia limit shouldn't be passed.

Limit test for Iron

Principle

- ❑ Limit test of Iron is based on the reaction of iron in ammonical solution with thioglycollic acid in presence of citric acid to form iron thioglycolate (Ferrous thioglycolate complex) which produces pale pink to deep reddish purple color in alkaline media.



- The color of the Ferrous thioglycolate complex fades in the presence of air due to oxidation.

Also, the color is destroyed in presence of oxidizing agent and strong alkalis. The purple color is developed only in alkaline media. So ammonia solution is used.

But,

- Ammonia reacts with iron, forms precipitate of ferrous hydroxide.
- Thus citric acid is used which prevents the precipitate of iron with Ammonia by forming a complex with iron as iron citrate.

Procedure

Test sample	Standard compound
Sample is dissolved in specific amount of water and then volume is made up to 40 ml	2 ml of standard solution of iron diluted with water upto 40 ml
Add 2 ml of 20 % w/v of citric acid (iron free)	Add 2 ml of 20 % w/v of citric acid (iron free)
Add 2 drops of thioglycollic acid	Add 2 drops of thioglycollic acid
Add ammonia to make the solution alkaline and adjust the volume to 50 ml	Add ammonia to make the solution alkaline and adjust the volume to 50 ml
Keep aside for 5 min	Keep aside for 5 min
Color developed is viewed vertically and compared with standard solution	Color developed is viewed vertically and compared with standard solution

Reasons

Citric Acid: Acts as a chelating agent that can bind iron ions, helping to stabilize the solution and prevent the precipitation of iron compounds.

Ammonia: Used to adjust the pH of the solution. It can create a more alkaline environment.

Thioglycolate: Functions as a reducing agent. It can reduce ferric ions (Fe^{3+}) to ferrous ions (Fe^{2+}).

Observation

- The purple color produced in sample solution should not be greater than standard solution.
- If purple color produced in sample solution is less than the standard solution, the sample will pass the limit test of iron and vice versa.

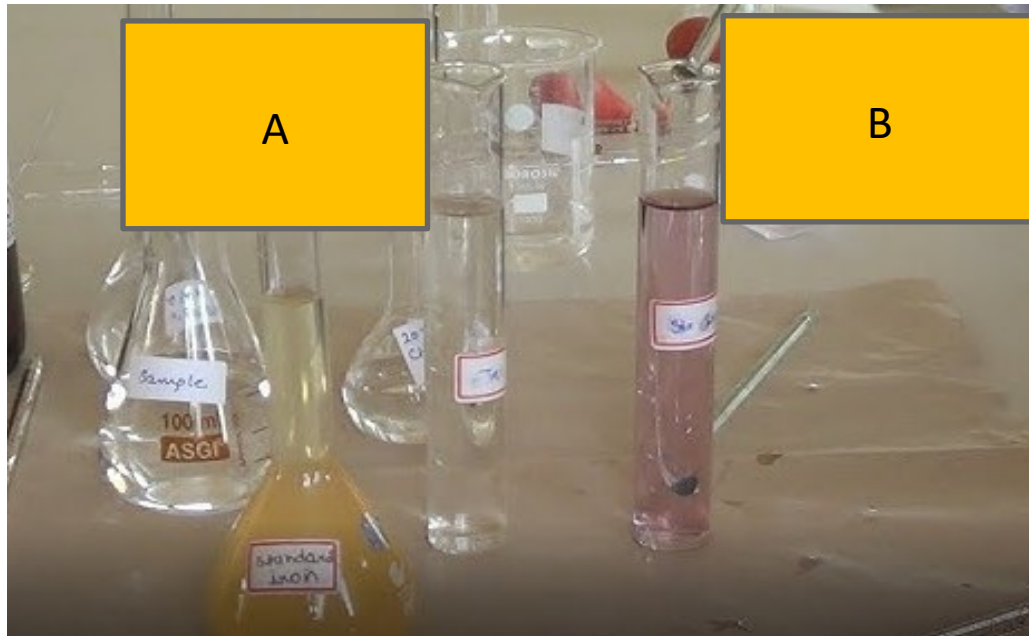


Figure: Solution A is the sample and ,Solution B is standard. Sample can pass the limit test of iron.

Limit test for Arsenic

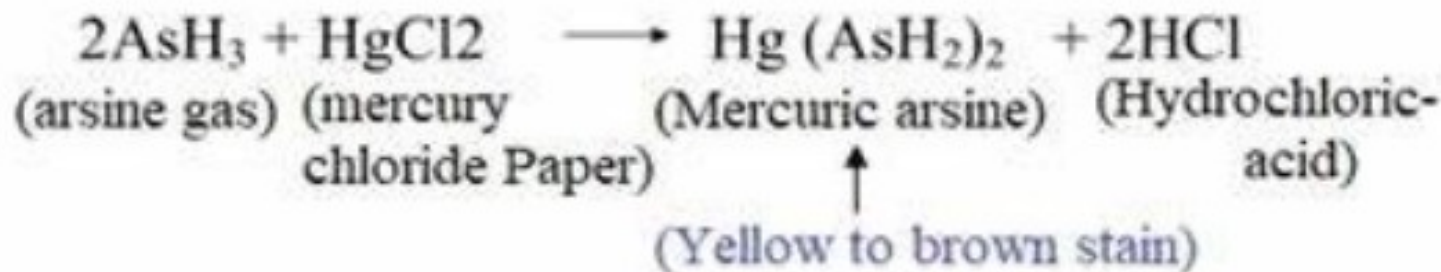
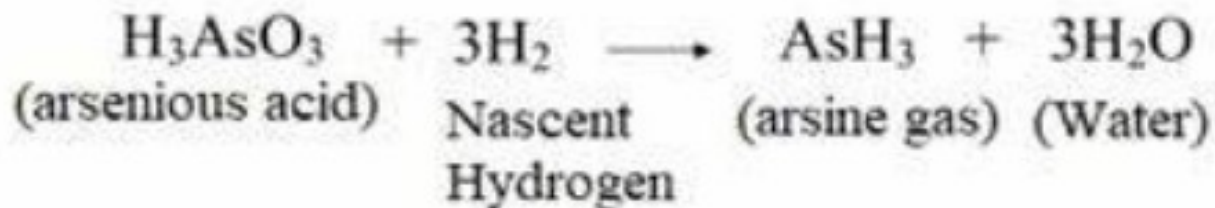
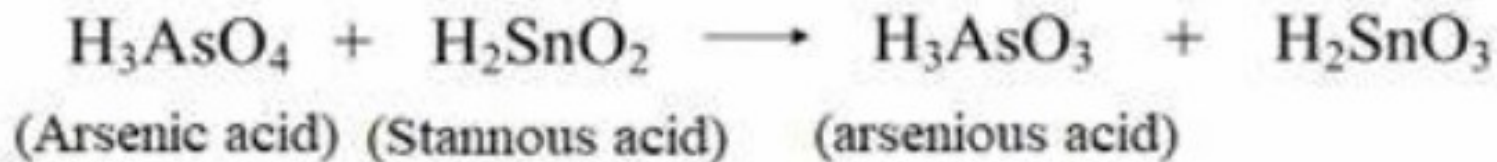
Arsenic is well known undesirable and harmful impurity which is present in medicinal substances.

Principle

- ❑ The limit test for arsenic is based on the reduction of the arsenic in the arsenious state to the arsine gas (AsH_3) with zinc and hydrochloric acid.
- ❑ If arsenic impurities present converted to arsine gas (AsH_3) which when contact with a mercuric chloride produces yellow stain.
- ❑ Zinc, potassium iodide and stannous chloride is used as a reducing agent.
- ❑ Hydrochloride acid is used to make the solution acidic.

Reaction

Arsenic-containing substances, when treated with dilute hydrochloric acid, yield arsenic acid.



- ❑ Arsenious acid is further reduced to arsine gas by nascent hydrogen and reacts with mercuric chloride paper to give yellow stain.
- ❖ The presence of stannous chloride and hydrochloric acid ensure rapid reaction between acid and potassium iodide and produce nascent hydrogen gas.
- ❖ The nascent hydrogen is produced by zinc and hydrochloric acid.

Comparison

- ❑ If the stain produced by the test is not deeper than the standard stain, then sample complies with the limit test for arsenic.

Procedure

SN	Standard	Test
1	A know amount of dilute arsenic solution is kept in the wide mouthed bottle of the apparatus.	The test solution is prepared by dissolving specific amount in water and stannated HCl (arsenic free) and kept in a wide mouthed bottle.
2	To this solution 1 gm of KI, 5 ml of stannous chloride acid solution and 10 gm of zinc is added (all this reagents must be arsenic free)	To this solution 1 gm of KI, 5 ml of stannous chloride acid solution and 10 gm of zinc is added (all this reagents must be arsenic free)
3	Keep the solution aside for 40 min	Keep the solution aside for 40 min
4	Compare the stain obtained on mercuric chloride paper with standard solution.	Compare the stain obtained on mercuric chloride paper with standard solution.

NON METALS LIMIT TEST

Limit test for Sulphate

Principle

Limit test of sulphate is based on the reaction of soluble sulphate with barium chloride in presence of dilute hydrochloric acid to form barium sulphate which appears as solid particles (turbidity) in the solution.



- ❑ Barium chloride is highly soluble in water , as a result it releases barium ion which will be attracted to sulfate ion.
- ❑ Barium sulfate precipitates out because its not water soluble ,then turbidity will appear as a marker.

Procedure

test sample	standard solution
solution is prepared as directed in the pharmacopoeia and transferred in Nessler cylinder	Take 1 ml of 0.1089% W/V solution of potassium sulphate in Nessler cylinder
Add 2 ml of dilute hydrochloric acid	Add 2 ml of dilute hydrochloric acid
Dilute to 45 ml in Nessler cylinder	Dilute to 45 ml in Nessler cylinder
Add 5 ml of barium sulphate reagent	Add 5 ml of barium sulphate reagent
Keep aside for 5 min observe turbidity	Keep aside for 5 min Observe the Turbidity

Barium sulphate reagent contains barium chloride, sulphate free alcohol and small of potassium sulphate. _____

Reasons

Hydrochloric acid helps to make the solution acidic to provide the protons that can trap the ions like carbonates ,sulfites and phosphates which will prevent false positive results.

Potassium sulphate is used to increase the sensitivity of the test by giving ionic concentration in the reagent. it releases more sulphate ion.

Alcohol helps to prevent super saturation and so produces a more uniform opalescence.

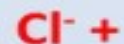
Observation

The turbidity produce in sample solution should not be greater than standard solution.

Limit Test for Chloride

Principle

- ❑ The chloride impurities in the substance are allowed to interact with silver nitrate solution in the presence of nitric acid.
- ❑ Nitric acid is added in the limit test of chloride to make solution acidic and helps silver chloride precipitate to make solution turbid at the end of process.
- ❑ This leads to the formation of silver chloride which is white in color. Since the quantity of chloride impurity is low, it forms an opalescence instead of a precipitate.



Soluble
chloride
present as
impurity



silver chloride



Procedure

Step s	Test solution	Standard solution
1	Dissolve sample in 10 ml of distilled water in a nessler cylinder labelled as “Test”	Take 1ml of standard sodium chloride solution into Nessler's cylinder and label it as “Standard”
2	Add 10ml of dilute nitric acids	Add 10ml of dilute nitric acid
3	Make up the solution to 50 ml mark using distilled water	Dilute the solution up to 50ml mark with distilled water
4	Add 1ml of 5% silver nitrate solution	Add 1ml of 5% of silver nitrate solution
5	Stir with glass rod and allow it stand for 5min	Stir with glass rods and wait for 5min.

Observation

- ❑ The opalescence produced in the test solution should not be greater than the standard solution to pass the limit test for chlorides for the particular sample.
- ❑ Finally, the opalescence is appeared , the formation of silver chloride which is white in color is the reason for opalescence formation.

In summary

The limit test for these atoms is essential for ensuring the quality and safety of pharmaceutical products because impurities above the acceptable limit can affect the efficacy, stability and physical properties of drug.

So controlling impurities is crucial to ensure that the drug remains effective ,toxicity free and stable throughout its shelf life.

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